

Topological Foundations: Jordan Curve Theorem

Formalization Notes

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Definition 1 (Homotopy and Nullhomotopy). *If f and f' are continuous maps of the space X into the space Y , we say that f is **homotopic** to f' if there is a continuous map $F : X \times I \rightarrow Y$ such that*

$$F(x, 0) = f(x) \quad \text{and} \quad F(x, 1) = f'(x)$$

*for each $x \in X$. (Here $I = [0, 1]$.) The map F is called a **homotopy** between f and f' .*

*If f is homotopic to f' , we write $f \simeq f'$. If $f \simeq f'$ and f' is a constant map, we say that f is **nullhomotopic**.*

Definition 2 (Path Homotopy). *Two paths f and f' mapping the interval $I = [0, 1]$ into X are said to be **path homotopic** if they have the same initial point x_0 and the same final point x_1 , and if there is a continuous map $F : I \times I \rightarrow X$ such that:*

$$F(s, 0) = f(s) \quad \text{and} \quad F(s, 1) = f'(s)$$

$$F(0, t) = x_0 \quad \text{and} \quad F(1, t) = x_1$$

*for each $s \in I$ and each $t \in I$. We call F a **path homotopy** between f and f' . If f is path homotopic to f' , we write $f \simeq_p f'$.*